

Principles of Microeconomics

Week 3 · Day 3 — Externalities, Public Goods, and the Commons

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When a market gets the quantity wrong because costs or benefits spill onto others.

A new reason markets fail

Until now, a competitive market gave the *efficient* outcome. Today it does not — because **prices leave something out**.

The theme

A choice can impose costs on, or give benefits to, people who are *not part of the deal*. The market then makes too much, or too little.

- **Negative externality:** a factory's pollution harms neighbours (too much made).
- **Public good:** a lighthouse helps every ship (too little provided — free riding).
- **The commons:** a shared fishery is over-fished (too much taken).

Learning goals — Day 3

By the end you can

- ① tell apart **private** and **social** cost, and find the *efficient* quantity under a negative externality;
- ② measure the **deadweight loss** and design a **corrective (Pigouvian) tax**;
- ③ explain the **free-rider problem** and classify goods by rivalry and excludability;
- ④ find the efficient amount of a **public good** and see why private provision falls short;
- ⑤ explain the **tragedy of the commons**.

Private cost vs. social cost

A competitive market for a chemical has demand and *private* marginal cost (the firms' own cost)

$$P = 100 - Q \quad (\text{marginal benefit}), \quad MPC = 10 + Q.$$

Each unit also creates pollution worth \$20 of harm to others — a **marginal external cost** $MEC = 20$. So the true cost to society is

$$MSC = MPC + MEC = 30 + Q.$$

The market ignores the harm

Firms produce where demand meets their *private* cost: $100 - Q = 10 + Q \Rightarrow \boxed{Q_{mkt} = 45}$.

Society counts the harm

The efficient quantity uses the *social* cost: $100 - Q = 30 + Q \Rightarrow \boxed{Q^* = 35}$.

Too much gets produced

The market makes too much

$Q_{mkt} = 45 > Q^* = 35$: each unit between 35 and 45 costs society more than anyone values it. The lost surplus is a triangle.

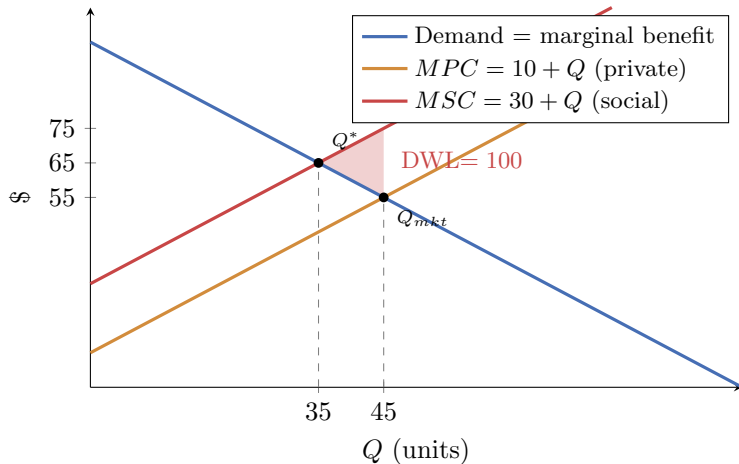
Deadweight loss

$$DWL = \frac{1}{2} \times \underbrace{(45 - 35)}_{\text{extra units}} \times \underbrace{20}_{\text{harm per unit}} = 100.$$

Reading the picture (next slide)

At $Q^* = 35$ the harm-inclusive cost just equals the benefit; by $Q = 45$ the gap between them has grown to \$20. The triangle adds up those gaps.

The externality — the picture



The market lands at $Q_{mkt} = 45$; efficiency wants $Q^* = 35$. The shaded triangle is the wasted surplus.

Fixing it: the corrective (Pigouvian) tax

Put a tax on each unit equal to the harm it causes: $t = MEC = 20$.

The tax makes the polluter “feel” the harm

The firm's cost becomes $MPC + t = 30 + Q$, which is exactly the social cost MSC . Now the market chooses

$$100 - Q = 30 + Q \Rightarrow Q = 35 = Q^*.$$

The \$100 deadweight loss disappears, and the tax raises revenue $20 \times 35 = 700$.

Other ways to fix it

- **Regulation / limits:** cap output or emissions directly.
- **Tradable permits:** issue a fixed number of pollution rights and let firms trade.
- **Bargaining (Coase):** with clear property rights and few parties, people can negotiate the fix themselves.

Four kinds of goods

Two questions decide how a good behaves in a market:

- **Rival?** Does my use reduce yours? (an apple: yes; a radio broadcast: no)
- **Excludable?** Can we stop non-payers? (a cinema: yes; street lighting: no)

	Excludable	Non-excludable
Rival	Private good (food, clothes)	Common resource (fish, pasture)
Non-rival	Club good (cinema, cable TV)	Public good (defence, street lights)

Today's troublemakers

Markets handle the top-left well. The two shaded cells fail: public goods (too little) and common resources (too much).

The free-rider problem

A **public good** is non-rival and non-excludable: once it exists, everyone enjoys it and no one can be shut out.

Why people “free-ride”

Because I benefit *whether or not I pay*, my best move is to let *others* pay and enjoy it for free. But everyone thinks the same way — so **too little** (or nothing) gets provided.

Familiar examples

National defence, clean air, public radio, a neighbourhood park, vaccination (herd immunity). All are hard to fund voluntarily — which is why governments often provide them.

How much public good is efficient?

A small town of two residents, A and B, chooses the size Q of a public good (say, park hectares). Their marginal benefits and the cost:

$$MB_A = 40 - Q, \quad MB_B = 60 - Q, \quad MC = 30.$$

Add the benefits *up* (vertically)

Because *everyone* enjoys the same park at once, society's benefit from one more hectare is the **sum** of what each person gets:

$$\text{total } MB = (40 - Q) + (60 - Q) = 100 - 2Q.$$

Efficient level: total $MB = MC$

$$100 - 2Q = 30 \implies \boxed{Q^* = 35}.$$

Society should build 35 — even though *neither* person alone values the 35th hectare at its \$30 cost.

Why voluntary contributions fall short

Left to themselves, each resident pays only up to where *their own* benefit equals the cost:

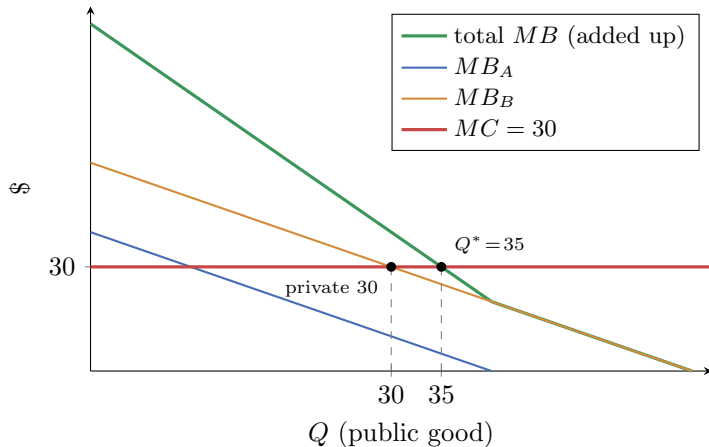
- Resident A: $40 - Q = 30 \Rightarrow Q = 10$.
- Resident B: $60 - Q = 30 \Rightarrow Q = 30$.

B (who values it most) provides 30; A sees 30 already there — where A's own benefit is only $40 - 30 = 10 < 30$ — and **contributes nothing**. A free-rides.

Under-provision

Private outcome $Q = 30 < Q^* = 35$. Each person ignores the benefit their contribution gives *everyone else*, so the park ends up too small.

The public good — the picture



Efficiency adds benefits *up* ($Q^* = 35$); voluntary giving stops at one person's benefit (30).

The tragedy of the commons

A common pasture is open to all villagers. Each cow yields a value that *falls* as the total herd N grows (the field gets crowded):

$$\text{value per cow} = 100 - N, \quad \text{cost per cow} = 20.$$

The efficient herd (if one owner managed it)

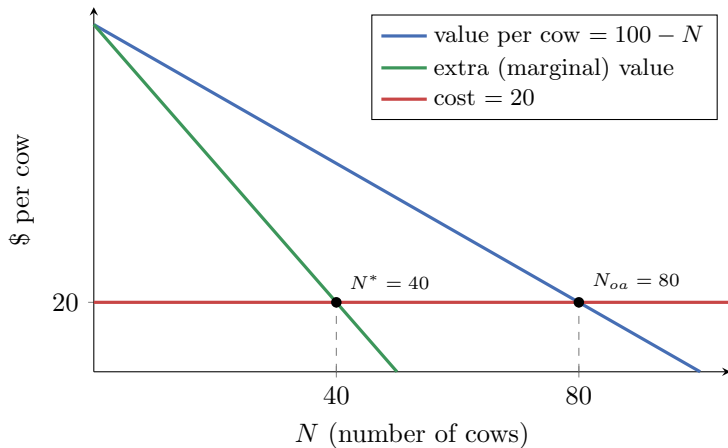
Add cows while the *extra* cow's value to the village beats its cost. The best herd is $N^* = 40$ (marginal value = cost).

Open access (everyone free to graze)

Each villager adds a cow while its *own* value covers the cost: $100 - N = 20 \Rightarrow N_{oa} = 80$. The herd **doubles**.

Each villager ignores that their extra cow makes *everyone else's* cows worth less — a negative externality on fellow users.

The commons — the picture



Efficiency sets the *extra* cow's value = cost ($N^* = 40$); open access keeps adding cows until the *average* value = cost ($N_{oa} = 80$).

Fixes: property rights (fencing), quotas / catch limits, use fees, or community rules.

One idea, three faces

	What spills over	Market's mistake / fix
Negative externality	a <i>cost</i> onto others	makes too much; tax at $t = MEC$
Public good	a <i>benefit</i> to all, non-excludable	provides too little; public provision
Common resource	a cost onto other <i>users</i>	over-used; property rights / limits

In every case, the price someone faces leaves out a cost or benefit borne by others. The fix is to make them *feel* it — through a tax, a right, or collective provision.

Day 3 — and week — checkpoint

Today you can

- find the efficient quantity under a negative externality and its deadweight loss;
- set a corrective tax $t = MEC$;
- find the efficient public good (add benefits up) and explain free-riding;
- explain the tragedy of the commons ($N_{oa} > N^*$).

The week in one line

Markets can fail through **strategic rivalry** (Days 1–2: game theory, repeated games) *and* through **spillovers** (Day 3: externalities, public goods, the commons) — and economics tells us both *why* and *how to fix it*.

In-class exercises — Day 3

Time for the in-class exercises.

- **Exercise 1** — a negative externality: efficient quantity, deadweight loss, and the corrective tax.
- **Exercise 2** — a public good (adding benefits up, free-riding) and a short tragedy-of-the-commons calculation.

Work in pairs; be ready to state the efficient quantity and the size of the market's error in each case.